

Lab #1

CoSc 20203
IBM System/360 Machine Instructions
Due: 5 Feb

Spring 2026

PROBLEM

The IBM System/360 series of mainframe computers was introduced in 1964 and led to IBM's dominance in the computer industry for 20 years. The instruction set had many formats, lengths, etc. We will consider only four instructions. An instruction in assembly language is written like

```
L      5,1000(0,12)
AR     5,1
```

The meaning of an instruction is not important for this assignment, but to partially satisfy your curiosity the first instruction says to load a value into register 5 from the memory location specified by 1000 plus the contents of register 12. The second instruction says to add the contents of register 1 to the contents of register 5.

In machine language, the instructions above look like

Binary	Hex
01011000 01010000 11000011 11101000	58 50 C3 E8
00011010 01010001	1A 51

A blank has been inserted between bytes to improve readability, but you should display the numbers without blanks.

The format of the instructions that we will consider is

Assembly Language	Machine Language
L R,D(X,B)	01011000 rrrrxxxx bbbbdddd dddddddd
A R,D(X,B)	01011010 rrrrxxxx bbbbdddd dddddddd
AR R,S	00011010 rrrrssss
ST R,D(X,B)	01010000 rrrrxxxx bbbbdddd dddddddd

Where “R”, “S”, “X”, and “B” are registers (0–15)
and “D” is the displacement (0–4095)

Notice the correspondence between uppercase letters in the second part of the assembly language and the lowercase letters in the machine language. For example, in

```
L      5,1000(0,12)
```

R is 5, D is 1000, X is 0, and B is 12. Thus,
r is 0101, d is 001111101000, x is 0000, and b is 1100.

The format of the assembly language must be exactly as shown. Blanks are legal only between the first part of the instruction and the second part. There must be at least one blank separating those parts but there can be more. In the assembly language, all numbers are decimal.

ASSIGNMENT

You will be given a Java application with a graphical user interface that allows a user to encode and decode IBM S/360 instructions. The user will enter a hex number or binary number and request that it be decoded into assembly language or enter an assembly language instruction and request that it be encoded into hex and binary representations. You must write the code for the actionPerformed methods for the “Encode” and “Decode” buttons. Your program needs to encode/decode only the four instructions shown above.

Display “ERROR” if the input cannot be encoded/decoded or there are formatting errors. There are many errors that you need to check for.

NOTE: This lab is intended to give you experience in using bitwise operators. **Do NOT use strings to create or manipulate the machine language instruction.**

To decode a machine language instruction, you must convert the input to an int and then use the bitwise operators to isolate each piece that corresponds to an element of the assembly language instruction.

To encode an assembly language instruction, you must start with a zero-valued int and then use the bitwise operators to set the relevant bits for each element of the assembly language instruction.